

Brutal Series - Paper B04 (Light, Respiration, Algebraic Expressions, Exponents & Powers)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. Two plane mirrors face each other at an angle of 60 degrees with a bead between them. The angle is then halved. How many images of the bead are seen after the change? [\[Light\]](#)
 - (A) 7
 - (B) 6
 - (C) 5
 - (D) 23
 - (E) 11
 - (F) 10
 - (G) 12

2. If $x = 3$ and $y = -2$, find the value of the expression $2x^2 - 3xy + y^2$. [\[Algebraic Expressions\]](#)
 - (A) 22
 - (B) 58
 - (C) 16
 - (D) 4
 - (E) 40
 - (F) 8

3. A person breathes 16 times each minute while resting. During heavy exercise the breathing rate triples and stays high for 5 minutes. How many breaths are taken in those 5 minutes? [\[Respiration in Organisms\]](#)
 - (A) 720
 - (B) 160
 - (C) 80
 - (D) 320
 - (E) 112
 - (F) 240
 - (G) 48

4. Simplify $(2^3 \times 2^4)$ divided by 2^5 and give the final number. [\[Exponents & Powers\]](#)
 - (A) 32
 - (B) 64
 - (C) 2
 - (D) 4
 - (E) 8
 - (F) 16

5. A girl stands 8 cm in front of a plane mirror. The mirror is then slid 3 cm farther away from her. What is the new distance between the girl and her image? [\[Light\]](#)
- (A) 19 cm
 - (B) 8 cm
 - (C) 14 cm
 - (D) 11 cm
 - (E) 16 cm
 - (F) 26 cm
 - (G) 22 cm
6. Subtract $(3a^2 - 5a + 7)$ from $(5a^2 + 2a - 3)$. What is the coefficient of a in the simplified result? [\[Algebraic Expressions\]](#)
- (A) -3
 - (B) -7
 - (C) 7
 - (D) -10
 - (E) 5
 - (F) 2
 - (G) 3
7. In one breath about 500 mL of air is taken in. If a child breathes 20 times every minute, how many litres of air are breathed in over 3 minutes? [\[Respiration in Organisms\]](#)
- (A) 10 L
 - (B) 30000 L
 - (C) 60 L
 - (D) 30 L
 - (E) 15 L
 - (F) 3 L
 - (G) 6 L
8. Find the value of $(-3)^4$ divided by $(-3)^2$ and state its sign. [\[Exponents & Powers\]](#)
- (A) -9
 - (B) -27
 - (C) 81
 - (D) 6
 - (E) 27
 - (F) 9 (positive)
 - (G) -81
9. A light ray meets a flat mirror so that the ray makes 35 degrees with the mirror surface. What angle separates the incoming ray from the reflected ray? [\[Light\]](#)
- (A) 130 degrees
 - (B) 70 degrees
 - (C) 145 degrees
 - (D) 90 degrees
 - (E) 110 degrees
 - (F) 35 degrees

10. The perimeter of a triangle is $(6x + 9)$. Two of its sides are $(2x + 4)$ and $(x + 1)$. Find the third side.

[\[Algebraic Expressions\]](#)

- (A) $3x + 5$
- (B) $5x + 4$
- (C) $3x - 4$
- (D) $3x + 4$
- (E) $3x + 14$
- (F) $9x + 14$

11. Yeast respires without oxygen. Glucose is broken down into a liquid and a gas that makes bread rise. Name the gas and the liquid product used in wine-making. [\[Respiration in Organisms\]](#)

- (A) Carbon dioxide; alcohol
- (B) Oxygen; alcohol
- (C) Carbon dioxide; glucose
- (D) Carbon dioxide; lactic acid
- (E) Carbon dioxide; water
- (F) Oxygen; water
- (G) Nitrogen; alcohol

12. Express the product 81×27 as a single power of 3. [\[Exponents & Powers\]](#)

- (A) 3^4
- (B) 3^{11}
- (C) 9^7
- (D) 27^4
- (E) 3^5
- (F) 3^7
- (G) 3^{12}

13. A wall clock seen in a plane mirror appears to read 3:40. What is the real time on the clock? [\[Light\]](#)

- (A) 8:20
- (B) 9:20
- (C) 9:40
- (D) 8:00
- (E) 8:40
- (F) 7:20
- (G) 3:40

14. Simplify $5(2x - 3) - 2(3x - 4)$ and then find its value when $x = 10$. [\[Algebraic Expressions\]](#)

- (A) -7
- (B) 33
- (C) 23
- (D) 7
- (E) 43
- (F) 47
- (G) 40

15. Counting one inhale and one exhale as a single breath, a sleeping person inhales 900 times in one hour. What is the breathing rate per minute? [\[Respiration in Organisms\]](#)

- (A) 60
- (B) 12
- (C) 15
- (D) 30
- (E) 9
- (F) 18
- (G) 45

16. The average distance from the Earth to the Sun is about 15,00,00,000 km. Write this in standard form $a \times 10^n$ where 1 is less than or equal to a , and a is less than 10. [\[Exponents & Powers\]](#)

- (A) 150×10^6 km
- (B) 1.5×10^6 km
- (C) 1.5×10^9 km
- (D) 0.15×10^9 km
- (E) 1.5×10^8 km
- (F) 1.5×10^7 km
- (G) 15×10^7 km

17. Two plane mirrors give exactly 5 images of a coin. To instead get 9 images, what angle must be set between the mirrors? [\[Light\]](#)

- (A) 20 degrees
- (B) 45 degrees
- (C) 36 degrees
- (D) 60 degrees
- (E) 72 degrees
- (F) 40 degrees

18. Add the three expressions $3x^2 - 2x + 5$, $-x^2 + 4x - 7$ and $2x^2 - x + 1$. Give the simplified sum. [\[Algebraic Expressions\]](#)

- (A) $4x^2 + 5x - 1$
- (B) $4x^2 + x + 1$
- (C) $4x^2 + x - 11$
- (D) $2x^2 + x - 1$
- (E) $6x^2 + x - 1$
- (F) $4x^2 - x - 1$
- (G) $4x^2 + x - 1$

19. During a sprint the muscles run short of oxygen and respire anaerobically, causing cramps. Which substance builds up in the muscles, and how does it differ from yeast's product? [\[Respiration in Organisms\]](#)

- (A) Alcohol; yeast makes lactic acid instead
- (B) Lactic acid; yeast makes alcohol and carbon dioxide instead
- (C) Water; yeast makes alcohol instead
- (D) Carbon dioxide; yeast makes water instead
- (E) Glucose; yeast makes alcohol instead
- (F) Lactic acid; yeast makes the same lactic acid
- (G) Lactic acid; yeast makes only oxygen

20. Simplify $(3^2)^3 \times 3^0$ and write the answer as a power of 3. [\[Exponents & Powers\]](#)

- (A) 1
- (B) 3^6
- (C) 3^5
- (D) 3^8
- (E) 3^7
- (F) 3^9
- (G) 3^2

21. White light is passed through a glass prism and splits into seven colours. Counting from the red end, which colour sits in the fourth place? [\[Light\]](#)

- (A) Indigo
- (B) Red
- (C) Green
- (D) Violet
- (E) Blue
- (F) Orange

22. If the value of $3x - 7$ is 11, find the value of $x^2 + 2x$. [\[Algebraic Expressions\]](#)

- (A) 48
- (B) 42
- (C) 8
- (D) 60
- (E) 15
- (F) 24
- (G) 36

23. Cockroaches and other insects breathe through tiny openings along the body that lead into fine air tubes. Name the openings and the tubes. [\[Respiration in Organisms\]](#)

- (A) Alveoli; spiracles
- (B) Spiracles; bronchi
- (C) Stomata; xylem
- (D) Spiracles; alveoli
- (E) Spiracles; tracheae
- (F) Skin pores; tracheae
- (G) Gills; trachea

24. Simplify $(7^5 \times 7^2)$ divided by $(7^3 \times 7^2)$ and give the final number. [\[Exponents & Powers\]](#)

- (A) 2401
- (B) 14
- (C) 49
- (D) 1
- (E) 343
- (F) 7
- (G) 16807

25. A vehicle's rear-view mirror always gives an erect but smaller image and a wide view of the road behind. Which type of mirror is it, and is the image real or virtual? [\[Light\]](#)

- (A) Plane mirror; real image
- (B) Concave mirror; virtual image
- (C) Convex mirror; inverted real image
- (D) Convex mirror; real image
- (E) Plane mirror; virtual image
- (F) Convex mirror; virtual image

26. From the sum of $(4p - 3q)$ and $(2p + 5q)$, subtract $(p - q)$. Simplify the result. [\[Algebraic Expressions\]](#)

- (A) $3p + 3q$
- (B) $7p + 3q$
- (C) $5p + q$
- (D) $7p + q$
- (E) $5p + 3q$
- (F) $5p - 3q$

27. Aerobic respiration of glucose uses oxygen and releases carbon dioxide, water and energy. In anaerobic respiration in our muscles, which products form and is more or less energy released? [\[Respiration in Organisms\]](#)

- (A) Lactic acid and energy; less energy released
- (B) Only water; the same energy released
- (C) Alcohol and carbon dioxide; less energy released
- (D) Lactic acid and energy; more energy released
- (E) Lactic acid and oxygen; less energy released
- (F) Carbon dioxide and water; more energy released

28. Express the number 1024 as a power of 2 and also as a power of 4. [\[Exponents & Powers\]](#)

- (A) 2^{10} and 4^6
- (B) 2^{10} and 4^5
- (C) 2^8 and 4^5
- (D) 2^9 and 4^5
- (E) 2^{11} and 4^5
- (F) 2^{12} and 4^6
- (G) 2^{10} and 4^4

29. A pin is placed between two plane mirrors set at 90 degrees. The angle is then changed to 45 degrees. By how many does the number of images increase? [\[Light\]](#)

- (A) 4
- (B) 8
- (C) 5
- (D) 7
- (E) 2
- (F) 3

30. A rectangle has length $(3x + 2)$ and breadth $(x + 4)$. Write its perimeter in simplest form. [\[Algebraic Expressions\]](#)

- (A) $4x + 12$
- (B) $8x + 12$
- (C) $4x + 6$
- (D) $8x + 24$
- (E) $8x + 8$
- (F) $6x + 12$

31. A patient's breathing rate rises from 18 to 30 breaths per minute. By what percentage did the breathing rate increase? [\[Respiration in Organisms\]](#)

- (A) 30%
- (B) 50%
- (C) about 66.7%
- (D) 12%
- (E) 40%
- (F) 60%
- (G) 166.7%

32. A sheet of paper 0.1 mm thick is folded so that its thickness doubles with each fold. After 10 folds, how many times thicker is it than one sheet, and how thick is it? [\[Exponents & Powers\]](#)

- (A) 512 times; 51.2 mm
- (B) 1000 times; 100 mm
- (C) 20 times; 2 mm
- (D) 100 times; 10 mm
- (E) 1024 times; 102.4 mm
- (F) 1024 times; 10.24 mm
- (G) 1024 times; 1024 mm

33. A ray of light strikes a plane mirror and the reflected ray makes an angle of 90 degrees with the incoming ray. What is the angle of incidence? [\[Light\]](#)

- (A) 30 degrees
- (B) 90 degrees
- (C) 180 degrees
- (D) 45 degrees
- (E) 0 degrees
- (F) 60 degrees
- (G) 22.5 degrees

34. Evaluate $a^2 - b^2$ for $a = 12$ and $b = 8$ by using the rule $a^2 - b^2 = (a + b)(a - b)$. [\[Algebraic Expressions\]](#)

- (A) 20
- (B) 16
- (C) 40
- (D) 208
- (E) 80
- (F) 400

35. Plants also respire. Through which tiny pores on a leaf do gases enter and leave, and which gas do plant cells take in during respiration? [\[Respiration in Organisms\]](#)

- (A) Lenticels; oxygen
- (B) Alveoli; oxygen
- (C) Stomata; oxygen
- (D) Spiracles; oxygen
- (E) Stomata; carbon dioxide
- (F) Stomata; hydrogen

36. Express 6,40,000 in standard form $a \times 10^n$. [\[Exponents & Powers\]](#)

- (A) 6.4×10^5
- (B) 0.64×10^6
- (C) 6.4×10^6
- (D) 64×10^4
- (E) 640×10^3
- (F) 6.4×10^4
- (G) 6.4×10^3

37. A man 180 cm tall wants to see his complete reflection in a plane mirror. What is the shortest mirror height that will let him see himself fully? [\[Light\]](#)

- (A) 60 cm
- (B) 100 cm
- (C) 45 cm
- (D) 180 cm
- (E) 135 cm
- (F) 90 cm

38. How many terms are there in the expression $7x^2y - 3xy + 4y^2 - 9$, and what is the constant term?

[\[Algebraic Expressions\]](#)

- (A) 4 terms; constant -9
- (B) 3 terms; constant 9
- (C) 4 terms; constant 4
- (D) 4 terms; constant 9
- (E) 3 terms; constant -9
- (F) 4 terms; constant 7
- (G) 5 terms; constant -9

39. Earthworms breathe through their moist skin. Why must this skin stay moist for breathing to work?

[\[Respiration in Organisms\]](#)

- (A) Dry skin produces its own oxygen faster
- (B) Moisture lets the worm breathe through lungs
- (C) Moisture blocks carbon dioxide from leaving
- (D) Moisture turns carbon dioxide into oxygen
- (E) Gases dissolve in the moisture and then pass through the skin
- (F) Water cools the worm so it needs less oxygen

40. Find the units (last) digit of 7^4 . [\[Exponents & Powers\]](#)

- (A) 7
- (B) 0
- (C) 1
- (D) 3
- (E) 9
- (F) 6

41. A candle stands 12 cm in front of a plane mirror. A screen is placed where the image appears to be. Will an image form on the screen, and how far apart are the candle and its image? [\[Light\]](#)

- (A) Yes; 6 cm apart
- (B) Yes; 24 cm apart
- (C) No; 12 cm apart
- (D) No; 48 cm apart
- (E) No; 6 cm apart
- (F) No image on the screen; 24 cm apart

42. Simplify the expression $2[3x - (2 - x)] + 5$. [\[Algebraic Expressions\]](#)

- (A) $8x + 3$
- (B) $8x - 1$
- (C) $4x + 1$
- (D) $8x + 5$
- (E) $6x - 1$
- (F) $6x + 1$
- (G) $8x + 1$

43. An athlete's resting breathing rate is 14 per minute. After a race it stays at 35 per minute for 4 minutes before returning to rest. How many EXTRA breaths were taken in those 4 minutes compared with resting?

[\[Respiration in Organisms\]](#)

- (A) 56
- (B) 21
- (C) 84
- (D) 70
- (E) 140
- (F) 196

44. Light travels about 3×10^5 km every second. How far does it travel in 100 seconds, written in standard form? [\[Exponents & Powers\]](#)

- (A) 3×10^8 km
- (B) 3×10^7 km
- (C) 3×10^3 km
- (D) 3×10^6 km
- (E) 30×10^5 km
- (F) 3×10^2 km

45. In the capital word CHOICE, how many letters look exactly the same as themselves when seen in a plane mirror (left-right reversal)? [\[Light\]](#)

- (A) 2
- (B) 0
- (C) 1
- (D) 4
- (E) 6
- (F) 3
- (G) 5

46. The age of a father is $5x$ years and his son is $(x + 4)$ years. The sum of their ages is 40 years. How old is the son? [\[Algebraic Expressions\]](#)

- (A) 16 years
- (B) 34 years
- (C) 30 years
- (D) 10 years
- (E) 6 years
- (F) 8 years
- (G) 12 years

47. When we breathe in, what does the diaphragm do, and how does the space inside the chest cavity change? [\[Respiration in Organisms\]](#)

- (A) It moves up and domes; the chest cavity increases
- (B) It moves up and domes; the chest cavity decreases
- (C) It moves down and flattens; the chest cavity decreases
- (D) It moves down; the chest cavity stays the same
- (E) It moves sideways; the chest cavity stays the same
- (F) It moves down and flattens; the chest cavity increases
- (G) It stays still; the chest cavity increases

48. Express $2^5 \times 5^5$ as a single power of 10 and give its value. [\[Exponents & Powers\]](#)

- (A) $10^6 = 1000000$
- (B) $10^5 = 1000000$
- (C) $10^5 = 100000$
- (D) $10^{25} = \text{very large}$
- (E) $10^5 = 10000$
- (F) $10^7 = 10000000$

49. Two plane mirrors are set at 120 degrees with an ant between them. How many images are seen, and is this more or fewer than at 90 degrees? [\[Light\]](#)

- (A) 2 images; fewer than at 90 degrees
- (B) 3 images; fewer than at 90 degrees
- (C) 1 image; fewer than at 90 degrees
- (D) 5 images; more than at 90 degrees
- (E) 3 images; more than at 90 degrees
- (F) 2 images; more than at 90 degrees
- (G) 4 images; fewer than at 90 degrees

50. Combine the like terms in $7ab - 3a + 2ab + 5a - ab$. [\[Algebraic Expressions\]](#)

- (A) $10ab + 2a$
- (B) $8ab + 2a$
- (C) $6ab + 2a$
- (D) $8ab + 3a$
- (E) $8ab + 8a$
- (F) $9ab + 2a$

51. A fish takes in oxygen dissolved in water using gills. A fish opens its mouth 80 times in 2 minutes. What is the rate per minute, and what organ does it pump water over? [\[Respiration in Organisms\]](#)

- (A) 40 per minute; spiracles
- (B) 40 per minute; lungs
- (C) 80 per minute; gills
- (D) 20 per minute; gills
- (E) 40 per minute; gills
- (F) 160 per minute; lungs

52. By what number should 3^4 be multiplied to give 3^7 ? [\[Exponents & Powers\]](#)

- (A) 27
- (B) 729
- (C) 9
- (D) 12
- (E) 6
- (F) 81

53. Light passes through a kaleidoscope built with two mirrors set at 36 degrees. How many images of a bead appear? [\[Light\]](#)

- (A) 11
- (B) 9
- (C) 6
- (D) 5
- (E) 10
- (F) 8
- (G) 12

54. What must be added to $(2x^2 - 3x + 1)$ to get $(5x^2 + x - 4)$? [\[Algebraic Expressions\]](#)

- (A) $7x^2 - 2x - 3$
- (B) $3x^2 + 4x + 5$
- (C) $3x^2 + 4x - 3$
- (D) $3x^2 - 4x - 5$
- (E) $3x^2 + 2x - 5$
- (F) $7x^2 + 4x - 5$
- (G) $3x^2 + 4x - 5$

55. Cells release energy by breaking down glucose during respiration. This usable energy is stored in a molecule whose short form is _____ before the cell spends it. [\[Respiration in Organisms\]](#)

- (A) ATP
- (B) RNA
- (C) WBC
- (D) DNA
- (E) CO₂
- (F) RBC
- (G) ADP

56. Simplify $2^0 + 3^0 + 4^0 + (5 + 6)^0$. [\[Exponents & Powers\]](#)

- (A) 0
- (B) 1
- (C) 18
- (D) 4
- (E) 8
- (F) 11

57. A boy raises his right hand in front of a plane mirror. Which hand does the image appear to raise, and what is this effect called? [\[Light\]](#)

- (A) The right hand; magnification
- (B) The right hand; lateral inversion
- (C) The left hand; total reflection
- (D) The left hand; lateral inversion
- (E) The right hand; reflection of light
- (F) The left hand; refraction
- (G) The left hand; dispersion

58. Multiply $(x + 3)$ by $(x - 5)$. State the constant term and the coefficient of x in the product. [\[Algebraic Expressions\]](#)

- (A) Constant +15; coefficient of x is +2
- (B) Constant -2; coefficient of x is -15
- (C) Constant -15; coefficient of x is -8
- (D) Constant -15; coefficient of x is -2
- (E) Constant +15; coefficient of x is -2
- (F) Constant -15; coefficient of x is +2
- (G) Constant -15; coefficient of x is +8

59. Lime water turns milky when we blow exhaled air through it. This happens because the air we breathe out is rich in which gas? [\[Respiration in Organisms\]](#)

- (A) Water vapour
- (B) Helium
- (C) Hydrogen
- (D) Oxygen
- (E) Carbon dioxide
- (F) Ozone
- (G) Nitrogen

60. Which is greater, 2^{10} or 10^3 , and by how much? [\[Exponents & Powers\]](#)

- (A) they are equal
- (B) 2^{10} is greater by 2
- (C) 2^{10} is greater by 24
- (D) 10^3 is greater by 24
- (E) 2^{10} is greater by 124
- (F) 2^{10} is greater by 1024
- (G) 10^3 is greater by 124

61. A car headlight sends out a strong parallel beam of light. Which type of mirror is used behind the bulb, and where is the bulb placed? [\[Light\]](#)

- (A) Plane mirror; bulb at the focus
- (B) Plane mirror; bulb at the centre
- (C) Concave mirror; bulb touching the mirror
- (D) Concave mirror; bulb very far away
- (E) Convex mirror; bulb at the focus
- (F) Convex mirror; bulb at the centre
- (G) Concave mirror; bulb at the focus

62. If $2x - 5y = 13$ and $x = 9$, find y , then evaluate $3y + 7$. [\[Algebraic Expressions\]](#)

- (A) 22
- (B) 1
- (C) 16
- (D) 34
- (E) 4
- (F) 10

63. The windpipe (trachea) carries air toward the lungs and then divides into two tubes, one entering each lung. What are these two tubes called? [\[Respiration in Organisms\]](#)

- (A) Gills
- (B) Alveoli
- (C) Spiracles
- (D) Capillaries
- (E) Bronchi
- (F) Bronchioles
- (G) Tracheae

64. Simplify $3^2 \times 3^3$ and give the answer as a power of 3 and as a number. [\[Exponents & Powers\]](#)

- (A) $3^5 = 15$
- (B) $6^5 = \text{large}$
- (C) $3^6 = 729$
- (D) $3^5 = 125$
- (E) $3^6 = 243$
- (F) $9^5 = \text{large}$
- (G) $3^5 = 243$

- 65.** A periscope is made with two plane mirrors inside a tube. At what angle is each mirror set to the tube, and how does the final ray compare with the original ray of light? [\[Light\]](#)
- (A) Each mirror at 30 degrees; the final ray reverses direction
 - (B) Each mirror at 60 degrees; the final ray is parallel to the original
 - (C) Each mirror at 45 degrees; the final ray is parallel to the original
 - (D) Each mirror at 45 degrees; the final ray reverses direction
 - (E) Each mirror at 90 degrees; the final ray is parallel to the original
 - (F) Each mirror at 90 degrees; the final ray is at right angles
- 66.** The pattern 3, 7, 11, 15, ... has n th term given by $4n - 1$. Which term of this pattern equals 95? [\[Algebraic Expressions\]](#)
- (A) the 19th term
 - (B) the 23rd term
 - (C) the 26th term
 - (D) the 22nd term
 - (E) the 21st term
 - (F) the 24th term
 - (G) the 25th term
- 67.** Tiny balloon-like air sacs in the lungs provide a huge surface for gas exchange. What are they called, and which gas passes from them into the blood? [\[Respiration in Organisms\]](#)
- (A) Alveoli; carbon dioxide
 - (B) Bronchioles; oxygen
 - (C) Bronchi; oxygen
 - (D) Alveoli; oxygen
 - (E) Alveoli; nitrogen
 - (F) Stomata; oxygen
- 68.** A bacterium divides into two every 20 minutes. Starting from a single bacterium, how many are there after 2 hours, written as a power of 2? [\[Exponents & Powers\]](#)
- (A) $2^{12} = 4096$
 - (B) $2^2 = 4$
 - (C) $2^8 = 256$
 - (D) $2^{120} = \text{huge}$
 - (E) $2^4 = 16$
 - (F) $2^6 = 64$
 - (G) $2^5 = 32$
- 69.** Two parallel plane mirrors face each other with a lit candle between them. How many images of the candle are formed? [\[Light\]](#)
- (A) 4
 - (B) 8
 - (C) 2
 - (D) Infinitely many
 - (E) 1
 - (F) 3
 - (G) 16

70. A square has each side of length $(2a + 1)$. Write its area as a trinomial (three-term expression). [\[Algebraic Expressions\]](#)

- (A) $4a^2 + 4a + 2$
- (B) $2a^2 + 4a + 1$
- (C) $4a^2 + 2a + 1$
- (D) $4a^2 + 4a$
- (E) $4a^2 + 1$
- (F) $2a^2 + 2a + 1$
- (G) $4a^2 + 4a + 1$

71. A tree carries out respiration day and night, but during the day it also does another process. Overall during daytime, which gas does the plant give out, and why? [\[Respiration in Organisms\]](#)

- (A) Carbon dioxide, because the plant only respire by day
- (B) Oxygen, because photosynthesis is faster than respiration
- (C) Carbon dioxide, because photosynthesis is faster than respiration
- (D) Oxygen, because the plant only respire by day
- (E) Carbon dioxide, because respiration stops in daylight
- (F) Oxygen, because respiration stops in daylight
- (G) Nitrogen, because both processes balance out

72. The population of a town is 1,25,000. Write it in standard form, and also as $125 \times 10^?$. [\[Exponents & Powers\]](#)

- (A) 1.25×10^4 and 125×10^3
- (B) 1.25×10^5 and 125×10^2
- (C) 1.25×10^6 and 125×10^3
- (D) 1.25×10^5 and 125×10^3
- (E) 12.5×10^4 and 125×10^3
- (F) 12.5×10^4 and 125×10^2
- (G) 1.25×10^5 and 125×10^4

73. An object and its image in a plane mirror are 30 cm apart. The object is then moved 4 cm closer to the mirror. What is the new distance between the object and its image? [\[Light\]](#)

- (A) 34 cm
- (B) 26 cm
- (C) 30 cm
- (D) 22 cm
- (E) 15 cm
- (F) 11 cm

74. Simplify $(3x^2 - 5x) - (x^2 - 5x + 6)$ and find its value when $x = -1$. [\[Algebraic Expressions\]](#)

- (A) -8
- (B) 8
- (C) 0
- (D) -4
- (E) 4
- (F) 2

- 75.** Frogs can breathe through both lungs and their moist skin. On land which surface is mainly used, and underwater which one is used? [\[Respiration in Organisms\]](#)
- (A) Lungs on land; gills underwater
 - (B) Lungs on land; skin underwater
 - (C) Lungs in both cases
 - (D) Gills on land; lungs underwater
 - (E) Gills on land; skin underwater
 - (F) Skin on land; lungs underwater
 - (G) Skin in both cases
- 76.** Compare 5^3 and 3^5 . Which is larger, and by how much? [\[Exponents & Powers\]](#)
- (A) 5^3 is larger by 118
 - (B) 3^5 is larger by 218
 - (C) 5^3 is larger by 18
 - (D) they are equal
 - (E) 3^5 is larger by 18
 - (F) 3^5 is larger by 118
- 77.** A ray of light falls along the normal (straight onto) a plane mirror. What are the angle of incidence and the angle of reflection? [\[Light\]](#)
- (A) 30 degrees and 30 degrees
 - (B) 0 degrees and 90 degrees
 - (C) 90 degrees and 0 degrees
 - (D) 45 degrees and 45 degrees
 - (E) 90 degrees and 90 degrees
 - (F) 0 degrees and 0 degrees
- 78.** Three consecutive whole numbers are n , $n + 1$ and $n + 2$. If their sum is 72, find the largest of the three numbers. [\[Algebraic Expressions\]](#)
- (A) 26
 - (B) 24
 - (C) 21
 - (D) 27
 - (E) 25
 - (F) 23
 - (G) 22
- 79.** A resting man takes in 6 litres of air per minute by breathing 12 times a minute at 0.5 litres per breath. If the depth stays the same but the rate rises to 20 breaths per minute, what is the new air intake per minute? [\[Respiration in Organisms\]](#)
- (A) 12 L
 - (B) 8 L
 - (C) 6 L
 - (D) 10 L
 - (E) 20 L
 - (F) 16 L
 - (G) 5 L

80. Find the value of $(4^3 \text{ divided by } 4^1)$ multiplied by 2. [\[Exponents & Powers\]](#)

- (A) 24
- (B) 128
- (C) 64
- (D) 16
- (E) 8
- (F) 32

81. Which one of the following statements about the image formed by a plane mirror is FALSE? [\[Light\]](#)

- (A) The image is as far behind the mirror as the object is in front
- (B) The image is erect (upright)
- (C) The image is virtual
- (D) The image is upright and same-sized
- (E) The image is real and can be caught on a screen
- (F) The image is laterally inverted
- (G) The image is the same size as the object

82. Find the coefficient of x in the simplified form of $3x(2x - 5) + 4(x + 2)$. [\[Algebraic Expressions\]](#)

- (A) 6
- (B) 11
- (C) 4
- (D) -11
- (E) -7
- (F) -15

83. During hard exercise a runner takes 30 breaths per minute, each breath 1.5 litres. What total volume of air, in litres, passes through the lungs in 6 minutes? [\[Respiration in Organisms\]](#)

- (A) 270 L
- (B) 180 L
- (C) 90 L
- (D) 45 L
- (E) 135 L
- (F) 540 L

84. Write the prime factorisation of 200 using exponents. [\[Exponents & Powers\]](#)

- (A) $2^3 \times 5^2$
- (B) $2^3 \times 5^3$
- (C) $2^3 \times 5$
- (D) $2^2 \times 5^3$
- (E) $2^2 \times 5^2$
- (F) 2×10^2
- (G) $2^4 \times 5^2$

- 85.** In a primary rainbow, the colours run from red to violet. Which colour appears on the top (outer) edge of the rainbow? [\[Light\]](#)
- (A) Indigo
 - (B) Blue
 - (C) Green
 - (D) Violet
 - (E) Orange
 - (F) Red
- 86.** The numerical coefficient of y in the term $-8xy$ is needed, and so is the degree of the term $6a^2b^3$. Give both. [\[Algebraic Expressions\]](#)
- (A) -8 and 5
 - (B) -8 and 6
 - (C) -1 and 5
 - (D) -8 and 2
 - (E) 8 and 5
 - (F) 8 and 6
 - (G) -8 and 3
- 87.** Why does bread dough rise when yeast is mixed in and the dough is kept warm? [\[Respiration in Organisms\]](#)
- (A) Yeast releases oxygen that forms bubbles
 - (B) Yeast respire and releases carbon dioxide gas that forms bubbles
 - (C) Yeast absorbs carbon dioxide and swells up
 - (D) Yeast turns flour directly into air pockets
 - (E) Heat alone boils the water into steam bubbles
 - (F) Yeast releases lactic acid that froths up
 - (G) Warmth melts the dough so it spreads out
- 88.** Standard form 4.5×10^4 represents which number, and how many zeros does that number have when written out fully? [\[Exponents & Powers\]](#)
- (A) 450; one zero
 - (B) 4500000; five zeros
 - (C) 4500; two zeros
 - (D) 45000; three zeros
 - (E) 450000; four zeros
 - (F) 45000; four zeros
 - (G) 45000; two zeros
- 89.** A concave mirror is used as a make-up or shaving mirror to give a larger image of the face. For this to work, where must the face be placed, and what kind of image forms? [\[Light\]](#)
- (A) Close to the mirror; an erect, enlarged, virtual image
 - (B) Far from the mirror; an erect, real image
 - (C) Close to the mirror; a diminished, virtual image
 - (D) Far from the mirror; an inverted, diminished image
 - (E) Touching the mirror; an inverted, real image
 - (F) Close to the mirror; an inverted, real image
 - (G) Far from the mirror; an erect, enlarged, virtual image

90. A taxi charges Rs $(8k + 30)$ for a trip of k kilometres. Find the charge for a 12 km trip, and the fixed charge that does not depend on distance. [\[Algebraic Expressions\]](#)

- (A) Rs 96; fixed Rs 8
- (B) Rs 120; fixed Rs 30
- (C) Rs 126; fixed Rs 0
- (D) Rs 156; fixed Rs 30
- (E) Rs 96; fixed Rs 30
- (F) Rs 126; fixed Rs 30
- (G) Rs 126; fixed Rs 8

91. Without oxygen, a muscle cell gets far less energy from one unit of glucose than it does with oxygen. This shows that aerobic respiration releases _____ energy than anaerobic respiration. [\[Respiration in Organisms\]](#)

- (A) the same
- (B) slightly less
- (C) no
- (D) more
- (E) equal
- (F) less
- (G) double less

92. Simplify $((-1)^7 + (-1)^8 + (-1)^9)$. [\[Exponents & Powers\]](#)

- (A) 1
- (B) 2
- (C) 3
- (D) -1
- (E) 0
- (F) -3
- (G) -2

93. Sunlight reflects off a small mirror and the reflected spot lands on a wall. If the mirror is rotated by 10 degrees, by how much does the reflected ray turn? [\[Light\]](#)

- (A) 20 degrees
- (B) 30 degrees
- (C) 10 degrees
- (D) 5 degrees
- (E) 0 degrees
- (F) 90 degrees
- (G) 40 degrees

94. If $p = 4$, evaluate the expression $p^3 - 2p^2 + 3p - 5$. [\[Algebraic Expressions\]](#)

- (A) 27
- (B) 43
- (C) 31
- (D) 35
- (E) 19
- (F) 39
- (G) 51

95. A person normally breathes 15 times per minute. While running, the rate doubles for 3 minutes and during that time each breath also doubles from 0.5 L to 1.0 L. By how many times does the air taken in per minute increase while running? [\[Respiration in Organisms\]](#)

- (A) 4 times
- (B) 1.5 times
- (C) 3 times
- (D) 16 times
- (E) 2 times
- (F) 6 times

96. Find the value of $(a^3 \times a^5)$ divided by a^2 when $a = 2$. [\[Exponents & Powers\]](#)

- (A) 512
- (B) 8
- (C) 32
- (D) 128
- (E) 16
- (F) 64
- (G) 256

97. A coin is placed between two plane mirrors and exactly 7 images are seen. What is the angle between the mirrors, and would a smaller angle give more or fewer images? [\[Light\]](#)

- (A) 45 degrees; a smaller angle would give fewer images
- (B) 60 degrees; a smaller angle would give more images
- (C) 90 degrees; a smaller angle would give more images
- (D) 30 degrees; a smaller angle would give fewer images
- (E) 72 degrees; a smaller angle would give more images
- (F) 45 degrees; the number would stay the same
- (G) 45 degrees; a smaller angle would give more images

98. The sum of the digits of a two-digit number is given by the expression $(t + u)$, where t and u are the tens and units digits. If $t = 2u$ and $t + u = 9$, find the original two-digit number. [\[Algebraic Expressions\]](#)

- (A) 54
- (B) 63
- (C) 72
- (D) 27
- (E) 18
- (F) 36
- (G) 45

99. A man's breathing rate is 18 per minute. He holds his breath for 40 seconds, then breathes normally for the rest of the minute. How many breaths does he take in that one minute (rounded down to a whole breath)? [\[Respiration in Organisms\]](#)

- (A) 12
- (B) 18
- (C) 5
- (D) 6
- (E) 9
- (F) 3
- (G) 8

100. Simplify $(10^6 \text{ divided by } 10^2)$ divided by 10^3 and write the answer as a power of 10 and as a number. [\[Exponents & Powers\]](#)

- (A) $10^3 = 1000$
- (B) $10^7 = \text{huge}$
- (C) $10^1 = 10$
- (D) $10^0 = 1$
- (E) $10^4 = 10000$
- (F) $10^{11} = \text{huge}$
- (G) $10^2 = 100$